

Algorithms and Data Structures

Problem Sheet 7

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62. Argue that since sorting n elements takes $\Omega(n \lg n)$ time in the worst case in the comparison model, any comparison-based algorithm for constructing a binary search tree from an arbitrary list of n elements takes $\Omega(n \lg n)$ time in the worst case.
63. Write the **TREE-PREDECESSOR** procedure for a binary search tree.
64. Professor Kilmer claims to have discovered a remarkable property of binary search trees. Suppose that the search for key k in a binary search tree ends up at a leaf. Consider three sets: A , the keys to the left of the search path; B , the keys on the search path; and C , the keys to the right of the search path. Professor Kilmer claims that any three keys $a \in A$, $b \in B$, and $c \in C$ must satisfy $a \leq b \leq c$. Give a smallest possible counterexample to the professor's claim.
65. Let T be a binary search tree whose keys are distinct, let x be a leaf node, and let y be its parent. Show that $y.key$ is either the smallest key in T larger than $x.key$ or the largest key in T smaller than $x.key$.
66. When **TREE-DELETE** calls **TRANSPLANT**, under what circumstances can the parameter v of **TRANSPLANT** be **NIL**?
67. Is the operation of deletion “commutative” in the sense that deleting x and then y from a binary search tree leaves the same tree as deleting y and then x ? Argue why it is or give a counterexample.
68. What is the largest possible number of internal nodes in a red-black tree with black-height k ? What is the smallest possible number?
69. Describe a red-black tree on n keys that realizes the largest possible ratio of red internal nodes to black internal nodes. What is this ratio? What tree has the smallest possible ratio, and what is the ratio?
70. Write pseudocode for **RIGHT-ROTATE**.
71. Show that any arbitrary n -node binary tree can be transformed into any other arbitrary n -node binary tree using $O(n)$ rotations.

Hint: Show that at most $n - 1$ rotations suffice to transform the tree into a right-going chain

72. The newly inserted node in **RB-INSERT** is set to the color red, which causes the problems with properties 2 and 4 of the RB-tree. If we would set it to black neither property would be violated. So why do we not set the new node to black instead of red?